

Ethan Tsui

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EDUCATION

University of South Florida, Tampa, FL |
Bachelor of Science in Mechanical Engineering

Expected Graduation: **May 2027**
GPA: 3.78/4.00

SKILLS & CERTIFICATIONS

- **CAD & Engineering:** SolidWorks (CSWA), AutoCAD, Fusion 360, MATLAB/Simulink
- **Prototyping & Testing:** FDM 3D Printing, Mechanical Assembly, Arduino, PCA9685, Servo Motors, Circuit Integration
- **Tools:** Git/GitHub, Microsoft Excel
- **Languages:** English (Native); Mandarin & Cantonese (Conversational)

PROJECTS

Project Atlas - 5-DOF Robotic Manipulator | Independent Project- GitHub Documentation June 2026 – September 2026

- Designed, fabricated, and tested a modular 5-DOF robotic manipulator using SolidWorks and FDM manufacturing from initial CAD through functional prototype.
- Diagnosed mechanical issues identified during testing and iteratively redesigned the joints, forearm, wrist, gripper, and bearing-supported base to improve range of motion, stiffness, manufacturability, and mechanical tolerances.
- Integrated 3 MG995 and 2 MG90S servos with an Arduino Nano and PCA9685 PWM controller for multi-axis manual control.
- Applied centre-of-mass and torque analysis to evaluate actuator loading and guide mechanical design decisions.

IAM3D Autonomous Rover Competition Project - Chassis Design Team

August 2025 – April 2026

- Designed and iterated a rover chassis and drivetrain to meet project requirements for stable traversal and material collection over sand terrain.
- Collaborated with a multidisciplinary engineering team to fabricate, troubleshoot, test, and refine the autonomous rover for the ASME IAM3D Competition at UCF.

WORK EXPERIENCE

Undergraduate Research Assistant

- USF Reality, Autonomy, and Robot Experience (RARE) Lab May 2026 – Present
- Collaborate with graduate researchers on Bloom robot mechanical improvements, maintaining SolidWorks CAD and design documentation through GitHub.
 - Redesigned the robot head to integrate a projector mounting system while maintaining manufacturability, component accessibility, and overall structural integrity.
 - Improved robotic arm reachability through iterative SolidWorks redesign and testing, increasing the robot's usable workspace.

Challenge Course Facilitator

- USF Recreation and Wellness September 2025 – Present
- Conduct pre-operation safety inspections of challenge-course grounds, elevated elements, and equipment to identify potential hazards and verify safe operating conditions before participant use.
 - Deliver safety briefings covering proper equipment use, course procedures, and participant expectations before activities begin.
 - Collaborate with a team of facilitators to supervise groups of up to 120 participants, enforce established safety procedures, and maintain safe course operations throughout activities.

LEADERSHIP AND EXTRACURRICULAR ACTIVITIES

Professional Development Chair | Tau Beta Pi Engineering Honor Society

Fall 2026 – Present

- Coordinate professional-development programming for USF's Florida Gamma chapter, including employer engagement, career-development events, and mock-interview programming.
- Collaborate with university staff, faculty, and industry professionals to develop networking and career opportunities for engineering students.

USF Boxing Club-Vice President

September 2023 - March 2024

- Co-led a funding pitch that secured \$10,000 for secure boxing equipment for members of the boxing club.
- Implemented structured training routines and promoted inclusivity to enhance member experience and engagement.

HONORS & CERTIFICATIONS

- Certified SolidWorks Associate (CSWA)
- SolidWorks Additive Manufacturing Associate
- SolidWorks Sustainability Associate
- Engineering Dean's List (Fall 2024, Spring 2026)
- Foundations Lab Robotics competition - 2nd Place in the Best Robot (Fall 2023) Mosaic